Biospectra : ISSN: 0973-7057, Vol. 20(2), September, 2025, pp. 75-78

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

Comparative evaluation of groundwater quality across selected villages of Ratu block, Ranchi, Jharkhand, India

Manoj Kumar Gupta* & Ajay Kumar Chaudhary

University Department of Zoology, DSPM University, Ranchi, Jharkhand, India

Received : 20th June, 2025 ; Revised : 24th July, 2025

DOI:-https://doi.org/10.5281/zenodo.19522946

Abstract- Groundwater is the main source of drinking and domestic water in many rural parts of India, but its quality is gradually changing due to increasing human activities and land use pressure. This study presents a comparative evaluation of groundwater quality across ten villages in Ratu Block, Ranchi, Jharkhand. Groundwater samples were collected from wells and tube wells and analyzed for major physico-chemical parameters such as pH, total dissolved solids (TDS), turbidity, total hardness, calcium, magnesium, sulphates, chlorides, and nitrates using standard methods. The results were compared with the drinking water standards of the Bureau of Indian Standards and the World Health Organization. The findings show clear variation in groundwater quality across the study area. The pH values ranged from 6.85 to 8.30, remaining within acceptable limits. TDS values varied between 390 and 465 mg/L, indicating moderate mineral content, while total hardness ranged from 195 to 230 mg/L, reflecting moderately hard to hard water conditions. Seasonal changes were also observed, with nitrate levels showing a slight increase during the monsoon due to surface runoff and agricultural inputs. A comparative assessment reveals that villages such as Bijuliya, Mariatu, Kantu, and Kathitad have relatively better water quality, whereas Ekaguri, Belangi, Makhmandro, Murgu, Simalia, and Brajpur show signs of deterioration and require treatment before consumption. The study underlines the need for regular monitoring and practical management measures to ensure safe and reliable groundwater for rural communities

Keywords: Groundwater, Water Quality Index, Physico-chemical parameters, Drinking water, Ratu Block

INTRODUCTION

Water is an essential natural resource that sustains life and supports a wide range of human activities, including drinking, agriculture, industry, and sanitation. Although approximately 75% of the Earth's surface is covered with water, only about 2.5% is freshwater, and an even smaller fraction is readily available for human consumption in the form of surface water and groundwater.[1] Groundwater, in particular, plays a crucial role in meeting drinking water requirements, especially in rural areas of developing countries like India, where nearly 80% of the rural population depends on it as a primary source of water.[2]

In recent decades, groundwater quality has deteriorated significantly due to rapid population growth, industrialization, urbanization, and intensified agricultural practices. Anthropogenic activities such as improper disposal of domestic waste, excessive use of chemical fertilizers and pesticides, and discharge of industrial effluents have introduced various contaminants into groundwater systems.[3] These pollutants alter the physico-chemical characteristics of water, making it unsafe for human consumption and posing serious health risks. Once groundwater is contaminated, its quality is difficult to restore, which highlights the importance of regular monitoring and management.[1]

*Corresponding author :
Phone : 7004715145
E-mail : pyaremanoj@yahoo.co.in

Assessment of groundwater quality typically involves the analysis of multiple physico-chemical parameters such as pH, total dissolved solids (TDS), turbidity, hardness, and major ions. However, interpreting large datasets containing numerous parameters can be complex and difficult for policymakers and the general public to understand. To address this issue, the Water Quality Index (WQI) has been developed as an effective and widely accepted tool for water quality assessment. WQI integrates multiple water quality parameters into a single numerical value that represents the overall status of water quality.[4] This approach simplifies data interpretation and facilitates better communication of water quality information for decision-making and resource management.[5]

Several studies have demonstrated the effectiveness of WQI in evaluating groundwater and surface water quality across different regions. It not only helps in classifying water quality into categories such as excellent, good, poor, or unsuitable but also aids in identifying key parameters responsible for water quality deterioration.[3]

The present study aims to evaluate the groundwater quality of Ratu Block, Ranchi, Jharkhand, using the Water Quality Index approach. The study also seeks to determine the suitability of groundwater for drinking purposes by comparing the observed values with standard guidelines prescribed by the Bureau of Indian Standards and the World Health Organization.

MATERIALS & METHODS

Study Area

The study was conducted in Ratu Block, Ranchi district, Jharkhand, India. The region is predominantly rural, with groundwater serving as the primary source of drinking and domestic water. Ten villages were selected for sampling: Bijuliya, Ekaguri, Mariatu, Kathitad, Simalia, Murgu, Belangi, Makhmandro, Kantu, and Brajpur.

Sample Collection

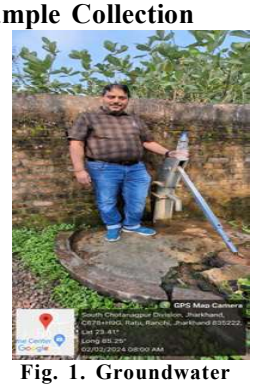


Fig. 1. Groundwater sampling at Kathitad



Fig. 2. Groundwater sampling at Near adiwasi school Kathitad

Groundwater samples were collected from wells and tube wells with depths greater than 50 feet to minimize contamination. Samples were collected in clean, pre-rinsed 1-liter polyethylene bottles. Proper labeling and preservation techniques were followed during transportation to the laboratory. Sampling was conducted during different seasons (pre-monsoon, monsoon, and post-monsoon) to assess temporal variations.

Physico-Chemical Analysis

The groundwater samples collected during the study were examined using standard procedures outlined by the American Public Health Association.[6] Various physico-chemical parameters were evaluated, including pH, which was determined by the electrometric method, and total dissolved solids (TDS). Turbidity was measured using suitable analytical methods, while total hardness, along with calcium and magnesium levels, was assessed through standard titrimetric techniques. The concentrations of chlorides, sulphates, and nitrates were also determined using established methods to evaluate the overall chemical characteristics of the groundwater.

Water Quality Index (WQI) Calculation

The Water Quality Index was calculated using the weighted arithmetic method:

$$WQI = \frac{\sum QiWi}{\sum Wi}$$

Where:

Qi represents the quality rating of each parameter

Wi represents the relative weight assigned to each parameter.

Water quality was classified as Excellent, Good, Poor, Very Poor, and Unsuitable based on WQI values.

RESULTS

The analysis of groundwater samples collected from different locations of Ratu Block reveals noticeable spatial variation in physico-chemical parameters, indicating the combined influence of natural geological conditions and anthropogenic activities.

The pH values of all samples ranged from 6.85 to 8.30, which fall within the permissible limits prescribed by the World Health Organization and the Bureau of Indian Standards.[1,2] This suggests that groundwater in the study area is neutral to slightly alkaline and does not pose any immediate risk related to acidity or alkalinity.

Total Dissolved Solids (TDS) showed moderate variation across the study area, with values ranging from 390 to 465 mg/L. Although most samples remained within

permissible limits, several locations approached the desirable limit of drinking water standards. This trend indicates moderate mineralization of groundwater, which may be associated with prolonged water-rock interaction as well as possible anthropogenic inputs such as domestic wastewater infiltration and agricultural runoff.[4]

Total hardness values ranged from 195 to 230 mg/L, indicating moderately hard to hard water conditions. The presence of calcium and magnesium ions is the primary contributor to hardness, which is typically derived from geological formations such as limestone and dolomite.[6] While hardness does not pose serious health risks, it can affect water palatability and lead to scaling in pipelines and household appliances.

Nitrate concentrations showed a slight increase during the monsoon season, suggesting the influence of surface runoff carrying fertilizers and organic matter into groundwater systems. Similar seasonal patterns have been reported in previous studies, where rainfall enhances nitrate leaching into aquifers.[5] Despite this increase, nitrate levels remained within permissible limits in most locations.

The concentrations of chlorides and sulphates were found to be within acceptable limits across all sampling sites, indicating minimal impact from industrial contamination or saline intrusion. This suggests that groundwater quality degradation in the study area is primarily influenced by natural and localized anthropogenic sources rather than large-scale industrial pollution.

Table 1 - Water Quality Index (WQI) Results

Village	pH	TDS (mg/L)	Hardness (mg/L)	WQI	Quality
Bijuliya	7.2	420	200	48.4	Good
Ekaguri	7	390	220	57.56	Poor
Mariatu	7.4	410	195	45.97	Good
Belangi	7.3	465	210	55.94	Poor
Makhmandro	6.9	430	230	61.86	Poor
Kantu	7.1	420	200	48.4	Good
Murgu	7.1	390	220	57.56	Poor
Kathitad	7.4	410	195	45.97	Good
Simalia	7.3	465	210	55.94	Poor
Brajpur	6.9	430	230	61.86	Poor

The computed WQI values ranged from 45.97 to 61.86, indicating that groundwater quality in the study area varies from good to poor. Villages such as Bijuliya, Mariatu, Kantu, and Kathitad fall within the good category, suggesting relatively safe water for drinking purposes. In contrast, Ekaguri, Belangi, Makhmandro, Murgu, Simalia, and Brajpur fall under the poor category, indicating the need for appropriate treatment before consumption.

The observed variation in WQI values highlights that TDS and hardness are the dominant factors influencing groundwater quality in the region, a trend that is consistent with findings reported in other regional studies.[3]

DISCUSSION

The results of the present study demonstrate that groundwater quality in Ratu Block is influenced by a combination of natural hydrogeological characteristics and human activities. The pH values remain within acceptable limits across all sampling locations, indicating stable chemical conditions and adequate buffering capacity within the aquifer system.[7,8]

Total Dissolved Solids (TDS) and total hardness emerge as the most critical parameters affecting groundwater quality. The observed TDS levels suggest moderate mineralization, which is commonly associated with prolonged interaction between groundwater and surrounding geological formations. Similar findings have been reported by Karbassi *et al.* (2011)[4], where elevated TDS levels were linked to natural geochemical processes. In addition, anthropogenic factors such as agricultural runoff, domestic wastewater infiltration, and improper waste disposal may further contribute to increased TDS concentrations.[3]

The hardness levels observed in the study area indicate the dominance of calcium and magnesium ions in groundwater, which are typically derived from carbonate-rich rocks such as limestone and dolomite. Comparable observations have been reported in other regions of India, where groundwater hardness is largely controlled by local geology.[6] Although hard water is not considered highly hazardous, prolonged consumption and domestic use can lead to operational and maintenance issues in water supply systems.

Seasonal variation plays a noticeable role in influencing groundwater quality, particularly in terms of nitrate concentration. The increase in nitrate levels during the monsoon season suggests enhanced leaching of fertilizers and organic matter into the groundwater system. This pattern is consistent with findings reported by Mukate *et al.* (2019)[5], who highlighted the role of rainfall in facilitating nitrate transport into aquifers. The presence of nitrates, even within permissible limits, indicates potential vulnerability of groundwater to agricultural contamination.

The Water Quality Index (WQI) provides a comprehensive understanding of groundwater quality by

integrating multiple parameters into a single value. The classification of several sampling locations under the "poor" category reflects localized deterioration of groundwater quality. Similar WQI-based studies conducted in different parts of India and abroad have also reported spatial variability in groundwater quality, often linked to differences in land use and population density.[3,5]

The variation observed among villages suggests that areas with higher agricultural activity and population pressure tend to exhibit relatively poorer water quality. This indicates that human activities play a significant role in influencing groundwater quality in addition to natural processes. The results emphasize the need for site-specific management strategies, including controlled use of fertilizers, improved sanitation practices, and protection of groundwater recharge zones.

The overall pattern observed in this study confirms that groundwater quality is dynamic and influenced by multiple interacting factors. Regular monitoring and the application of tools such as WQI can provide valuable insights for effective groundwater management and policy planning.

CONCLUSION

The present study provides a detailed assessment of groundwater quality in Ratu Block, Ranchi, using the Water Quality Index (WQI) approach. The analysis of physico-chemical parameters reveals that groundwater quality varies considerably across the selected villages.

The pH values of all samples ranged from 6.85 to 8.30, remaining within the permissible limits prescribed by BIS and WHO standards, indicating that acidity and alkalinity are not major concerns. Total Dissolved Solids (TDS) values ranged approximately between 390 and 465 mg/L, with several locations approaching the desirable limit, suggesting moderate mineralization of groundwater. Total hardness values ranged from 195 to 230 mg/L, indicating moderately hard to hard water conditions, primarily due to the presence of calcium and magnesium ions.

The computed Water Quality Index (WQI) values ranged from 45.97 to 61.86 across the study area. Villages such as Bijuliya, Mariatu, Kantu, and Kathitad fall under the “good” category (WQI < 50), indicating relatively safe water for drinking purposes. In contrast, Ekaguri, Belangi, Makhmandro, Murgu, Simalia, and Brajpur fall under the “poor” category (WQI > 50), suggesting that water from these locations requires treatment before consumption.

The study clearly identifies TDS and total hardness as the dominant factors influencing groundwater quality in the region. Seasonal variations, particularly during the monsoon, also contribute to slight increases in nitrate levels due to surface runoff and agricultural activities.

These findings indicate that while groundwater in some parts of Ratu Block is suitable for drinking, a significant portion requires appropriate treatment and management. Continuous monitoring, along with the implementation of water treatment methods and sustainable groundwater management practices, is essential to ensure safe and reliable water supply for the local population.

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